

COUNTING

CSE 2213 – Discrete Mathematics

BASIC COUNTING PRINCIPLES

Product rule

Sum rule

THE PRODUCT RULE

PRODUCT RULE

If a task can be divided into two subtasks, and these subtasks can be individually completed in n_1 and n_2 ways respectively, then there are $n_1 n_2$ ways to complete that task

In general, if a task can be divided into m subtasks, and they can be individually completed in $n_1, n_2, \dots, n_m$ ways respectively, then the total number of ways to complete the task is –

$$\prod_{i=1}^m n_i$$

BIRTHDAY TREAT

Suppose, you want to give birthday treat to your friends in a restaurant. The restaurant offers 3 main course (say, Biryani, Pizza, Pasta) and 2 types of cold drinks (Pepsi, 7Up). Your friend can choose one main course and one cold drink from the given options. How many different choices does your friend have to select the meal ?

You have to select two items from the menu.

One from the main course, for that you have total three choices.



Another one from the cold drinks, for that you have two options.



If you choose Biryani, then you can take pepsi or 7up with it. Therefore, you have two different choices.



+



+



If you choose Pizza, then you can take pepsi or 7up with it. Therefore, you have two choices.



+



+



If you choose Pasta, then you can take pepsi or 7up with it. Therefore, you have two choices.



+



+



SOLVING A PROBLEM USING PRODUCT RULE

Step one: Identify the subtasks

- The subtasks **SHOULD** be done **ONE AFTER THE OTHER** to make the whole task possible
- If you think about it, you will notice that the subtasks are going to be linked with an “AND” relationship

In our birthday treat problem, there are mainly two sub tasks:

➤ Select food item from the main course

AND

➤ Select cold drinks from the available cold drinks.

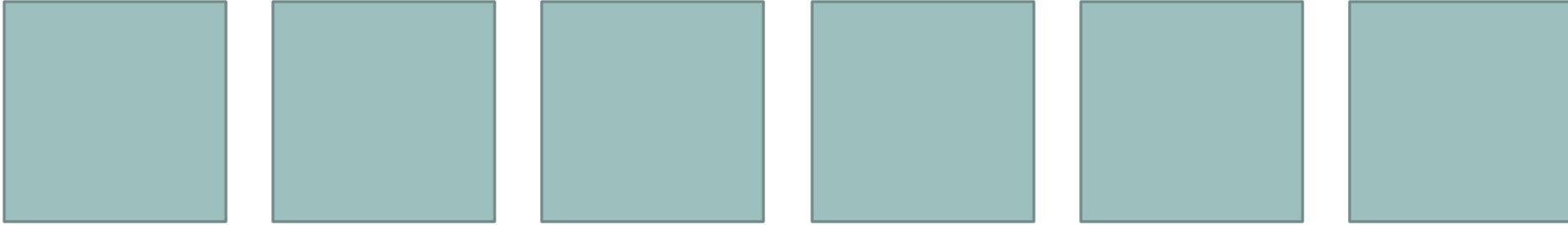
Step two: Count the possible ways for each subtask

- Select food item from the main course: There are 3 ways to choose it.
- Select cold drink from the available cold drinks: There are 2 ways to choose it.

Step three: Multiply the possible ways : $3 \times 2 = 6$ ways you can select the meal.

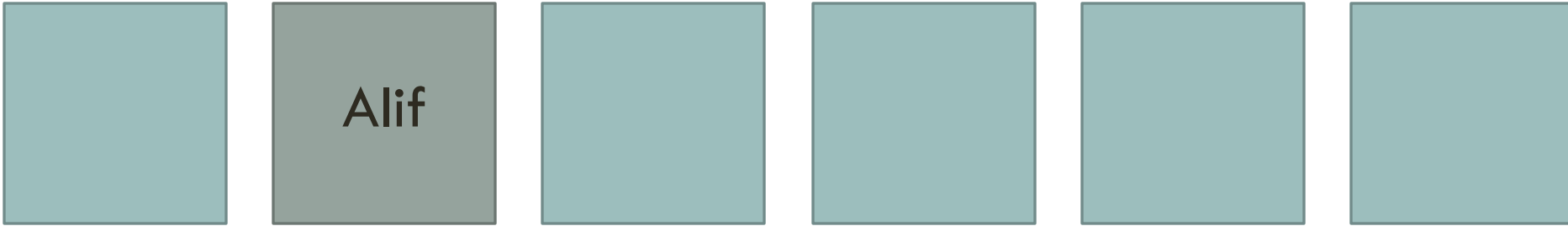
EXAMPLE

There are twelve empty rooms in the office of Acme Softwares, inc. Alif and Laila joins the company. How many ways can they be assigned a room each from these twelve?

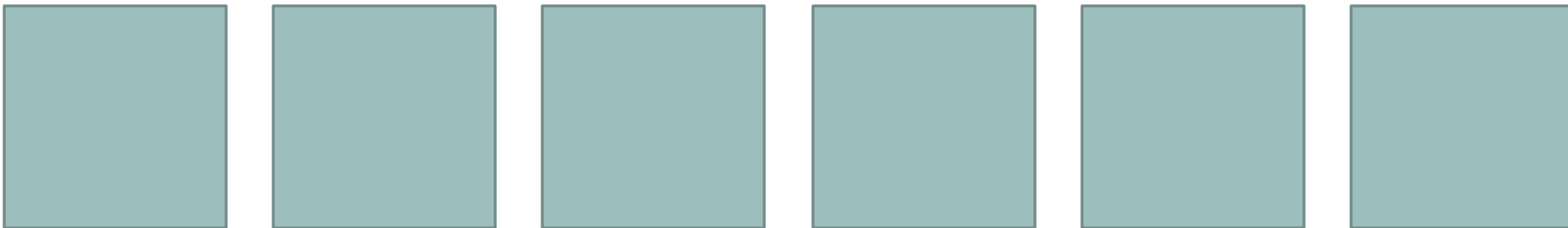


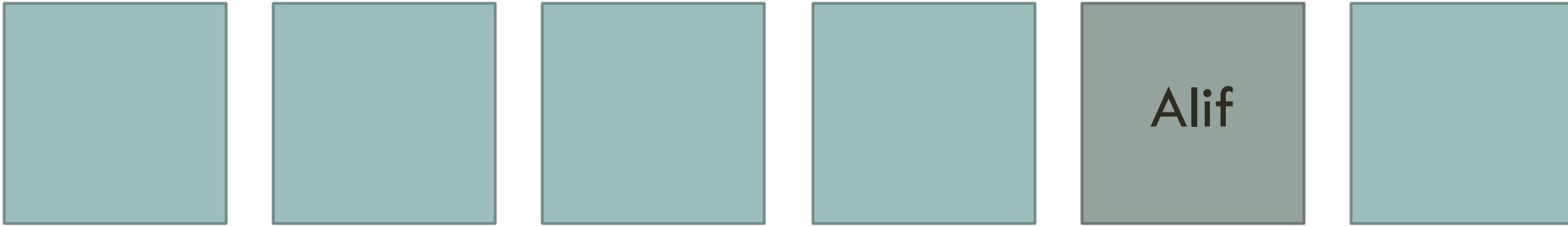
Total choice for Alif = 12



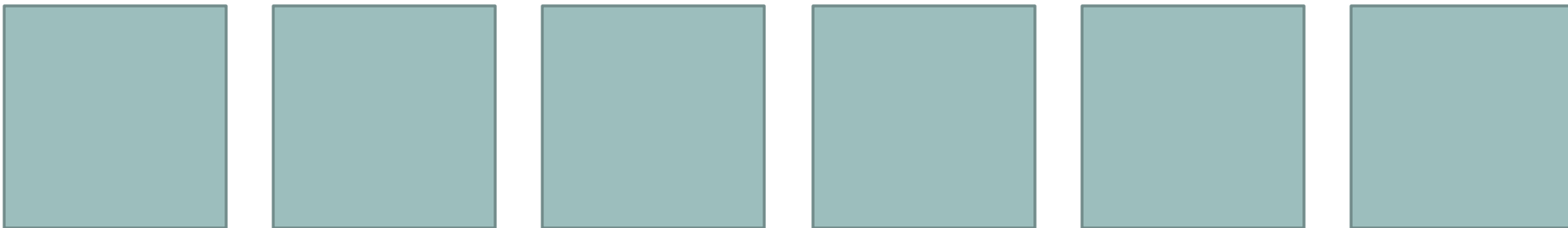


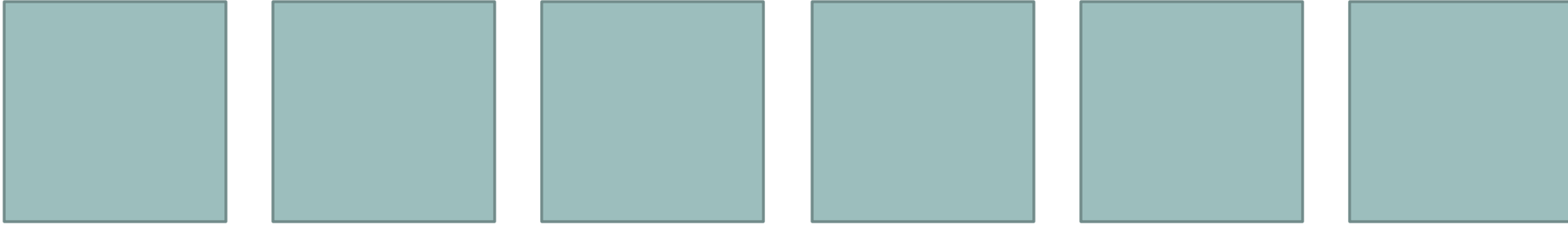
Total choice for Laila = 11





Total choice for Laila = 11





Total choice for Laila = 11





For every possible room assignment for Alif,
there are 11 choices for Laila

Total no. of assignments = $12 \times 11 = 132$



EXAMPLE

What is the total number of bit strings of length 7?

BIT STRING

A string containing only 0's and 1's

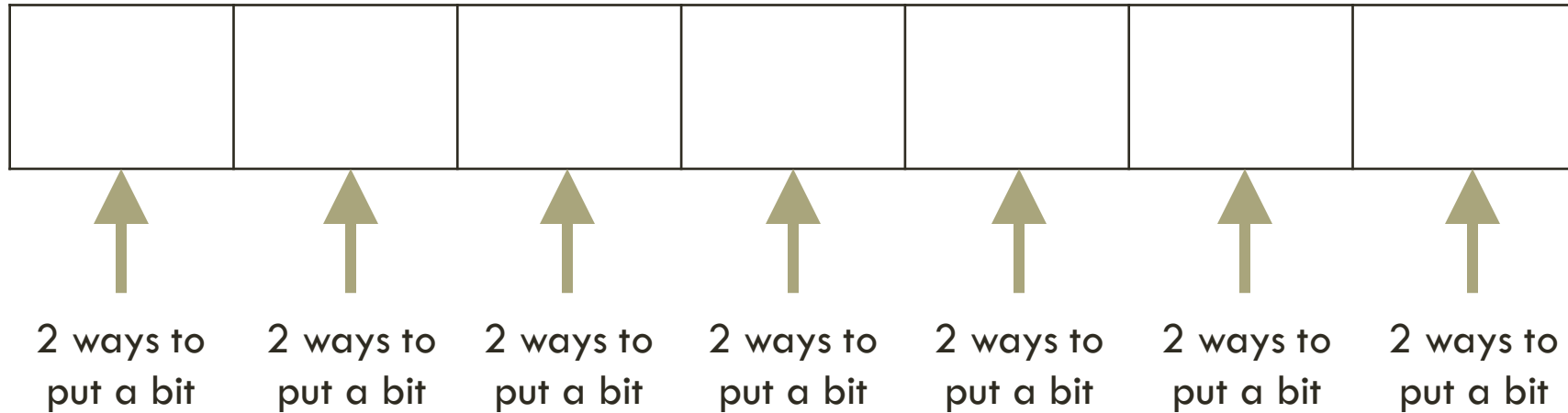
A bit string of length 7 has 7 places to put a bit in

--	--	--	--	--	--	--

How many such bit strings are there?

PAUSE THE VIDEO, AND TRY YOURSELF

The problem has 7 subtasks – putting a bit in each of the 7 places



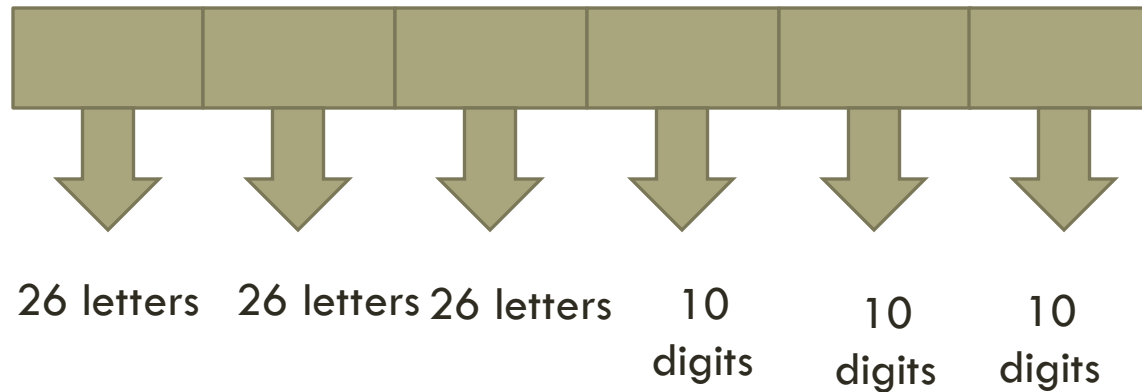
$$\text{Total number of ways} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$$

EXAMPLE

How many different license plates are available if each plate contains a sequence of three lowercase letters followed by three digits (and no sequences of letters are prohibited)?

This problem has 6 sub tasks-putting three letters in the first three places and then three digits in the remaining the three places.

We have 26 letters in English and for digit (0-9)= total 10 different digits



$$\text{Total number of license plates} = 26 \times 26 \times 26 \times 10 \times 10 \times 10 = 26^3 \times 10^3$$

THE SUM RULE

THE SUM RULE

If a task can be done in one of n_1 ways or in one of n_2 ways, where none of the two types are the same, then there are $n_1 + n_2$ ways to do that task

In general, if a task can be done in m options each of which has $n_1, n_2, \dots, n_m$ ways respectively, then the total number of ways to do that task is –

$$\sum_{i=1}^m n_i$$

LET'S TRAVEL FROM DHAKA TO CHATTOGRAM

We can travel from Dhaka to Chattogram in bus, train or by air

How many buses are there?

- Shohag, Green line, Hanif

How many trains are there?

- Mahanagar, Subarna, Sonar Bangla, Turna

How many flights are there?

- Novoair, US-Bangla

So how many ways we can travel?

TRAVEL EXAMPLE

Buses: Shohag, Green Line, Hanif → 3

Trains: Mahanagar, Subarna, Sonar Bangla, Turna → 4

Flights: Novoair, US-Bangla → 2

Total = 3 + 4 + 2 = 9

SOLVING A PROBLEM USING SUM RULE

Step one: Identify the different options of doing the **SAME** task

- The different options are going to be mutually exclusive
- If you think about it, you will notice that in order to carry out the task, you will choose one option or the other
- That is, the different options are going to be connected with an **OR** relationship

Step two: Count the possible ways for each options

Step three: Add the possible ways

EXAMPLE

Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

SOLUTION

1. Identify the different options:

Option 1: To choose representative from students who are mathematics major.

Option 2: To choose representative from mathematics faculty.

2. Count the possible ways for each options

Option 1: 83 students

Option 2: 37 Faculty members

3. Add the possible ways

$83 + 37 = 120$ ways